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Course: Applied cryptography.
Assignment

1. Fermat's Theorem:

Compute $92^{46} \bmod 47$ using Fermat's Little Theorem.

$$\Rightarrow 92^{46} \bmod 47$$

$$\Rightarrow 91 \equiv 44 \equiv -3 \pmod{47}$$

$$\Rightarrow a^{46} \equiv 1 \pmod{47}$$

$$\Rightarrow \gcd(91, 47) = 1$$

$$91^{46} \equiv 1 \pmod{47} \Rightarrow 1$$

2. Euler's Theorem:

$$106 = 2 \times 53$$

$$\phi(106) \Rightarrow 106 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{53}\right)$$

$$\Rightarrow 106 \times \frac{1}{2} \times \frac{52}{53} = 52$$

$$\Rightarrow 7^{52} \pmod{106}$$

$$\gcd(7, 106) = 1$$

$$7^{52} \equiv 1 \pmod{106}$$

$$\phi(106) = 52$$
$$7^{52} \bmod 106 = 1.$$

3) RSA: $p=13, q=19, e=7, M=8$

$$n=pq \Rightarrow 13 \times 19 = 247$$

$$\phi(n) \Rightarrow (13-1)(19-1) \Rightarrow 12 \times 18 = 216$$

To find d :

$$e \times d \equiv 1 \pmod{216}$$

$$7d \equiv 1 \pmod{216}$$

$$d=31$$

public key $\Rightarrow (e, n) \Rightarrow (7, 247)$

$(d, n) \Rightarrow (31, 247)$

encryption:

$$\Rightarrow 8^7 \pmod{247}$$

$$8^2 \Rightarrow 64$$

$$8^4 \Rightarrow 4096 \equiv 144$$

$$8^7 \Rightarrow 8^4 \times 8^2 \times 8$$

$$\Rightarrow 144 \times 64 \times 8$$

$$\Rightarrow 77 \times 8 = 616 \equiv 122$$

$$C = 122$$

Decryption:

$$M = C^{31} \pmod{247}$$

$$122^{31} \equiv 8 \pmod{247}$$

$$\Rightarrow 8$$

Diffie Hellman:

$$q=23, a=5, x_A=3, x_B=10$$

$$y_A = 5^3 \pmod{23}$$

$$125 \equiv 10 \quad \boxed{y_A = 10}$$

$$y_B = 5^{10} \pmod{23} \rightarrow 5^{10} \equiv 9 \quad \boxed{y_B = 9}$$

$$k = 9^3 \pmod{23}$$

$$9^2 = 81 \equiv 12$$

$$12 \times 9 = 108 \equiv 16$$

$$10^{10} \pmod{23} = 16$$

$$\neq 16$$

b). ElGamal:

$$q=23, a=5, x_A=9, k=3, M=8$$

public key: $q=23, a=5, x_A=9, k=3$
 $M=8$

$$y_A = 5^9 \pmod{23}$$

$$\boxed{y_A = 11}$$

Encryption:

$$C_1 = 5^3 \pmod{23} = 10$$

$$k = 11^3 \pmod{23}$$

$$= 11^2 = 121 \equiv 6$$

$$6 \times 11 = 66 \equiv 20$$

$$\boxed{k=20}$$

$$C_2: C_2 = 8 \times 20 \pmod{23}$$

$$160 \equiv 22$$

$$(10, 22)$$

Decryption:

$$k = C_1^{x_A} = 10^9 \pmod{23}$$

$$k=20$$

$$20^{-1} = 15$$

$$20 \times 15 = 30 \equiv 1 \pmod{23}$$

$$M = 22 \times 15 \pmod{23}$$

$$330 \equiv 8$$

$$\Rightarrow 8$$

Miller Rabin : 193 is primary key
using base $a=2$.

Step 1 $\rightarrow n-1 \Rightarrow 2^k \times q$

$$193-1 \Rightarrow 192$$

$$192 / 2^1 \rightarrow 96$$

$$192 / 2^2 \rightarrow 48$$

$$192 / 2^3 \rightarrow 24$$

$$192 / 2^4 \rightarrow 12$$

$$192 / 2^5 \rightarrow 6$$

$$192 / 2^6 \rightarrow 3$$

$$192 / 2^7 \rightarrow 1.5$$

$$k=6 ; q=3$$

Step 2: $a=2$

Step 3: $b_0 = a^q \pmod{n}$

$$\rightarrow a^3 \pmod{193}$$

$$\rightarrow 2^3 \pmod{193}$$

$$\rightarrow 2^3 \pmod{193}$$

$$\rightarrow 8$$

$$b_2 \rightarrow b_0^2 \pmod{n}$$

$$\rightarrow 8^2 \pmod{193} \rightarrow 64$$

$$b_1 \rightarrow 8^2 \bmod 193$$

$$\rightarrow 64$$

$$b_2 \rightarrow 4096 \bmod 193$$

$$\rightarrow \cancel{1376} \rightarrow 43$$

$$b_3 \rightarrow 6^2 \bmod 193$$

$$\rightarrow 1849 \bmod 193$$

$$\rightarrow 112$$

$$b_5 \rightarrow 6^2 \bmod 193$$

$$\rightarrow 12544 \bmod 193$$

$$\rightarrow 192$$

$\therefore 193$ passes the test

$\therefore$ It is probably prime.

Amu

Diffie Hellman $q=23$, $a=5$, $x_A=3$, $x_B=10$
find y_A , y_B and shared key.

Step 1: $y_A = a^{x_A} \bmod q$
 $\Rightarrow 5^3 \bmod 23$
 $\Rightarrow 125 \bmod 23 \Rightarrow 10$

Step 2: $y_B = a^{x_B} \bmod q$
 $\Rightarrow 5^{10} \bmod 23$

$$5^2 \Rightarrow 25 \equiv 2 \pmod{23}$$

$$5^4 \Rightarrow 2^2 \Rightarrow 4 \pmod{23}$$

$$5^8 = 4^2 = 16 \pmod{23}$$

$$\Rightarrow 16 \times 2 = 32 \Rightarrow 32 \bmod 23 = 9$$

$$K = y_B^{x_A} \bmod q = 9^3 \bmod 23$$

$$9^2 = 81 \equiv 12 \pmod{23}$$

$$9^3 = 12 \times 9 = 108$$

$$108 \bmod 23 = 16$$

$$\Rightarrow 16$$

$$k = y_A^{x_B} \bmod q$$

$$\Rightarrow 10^{16} \bmod 23$$

$$\Rightarrow 10^2 = 100 \equiv 8 \pmod{23}$$

$$\Rightarrow 10^4 = 8^2 = 64 \equiv 18 \pmod{23}$$

$$\Rightarrow 10^8 = 18^2 = 324 \equiv 2 \pmod{23}$$

$$\Rightarrow 16 \pmod{23}$$
